SCS 1306 - Linear Algebra

Tutorial 04

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01) The inverse of the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 3 \end{bmatrix}$ is the matrix $A^{-1} = \begin{bmatrix} a & 1 & -1 \\ 1 & b & 1 \\ -1 & 1 & 0 \end{bmatrix}$.

Determine the values of a and b.

02) Use elementary row operations to transform

(a) the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 3 & 1 & 2 \end{bmatrix}$ into a *upper triangular* matrix;

(b) the matrix $B = \begin{bmatrix} 1 & 3 & 3 \\ 2 & 4 & 10 \\ 3 & 8 & 4 \end{bmatrix}$ into a *unit* matrix.

03) Let $A = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$ and $B = \begin{pmatrix} 1 & \tan\theta \\ 0 & 1 \end{pmatrix}$.

Show that A and B are row equivalent.

04) Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$.

Find (a) A^{-1}, (b) A^5 and (c) A^{-5}.

05) Using elementary transformations find the inverse of each of the following matrices:

(a) $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$;

(b) $B = \begin{bmatrix} 2 & -6 & -2 & -3 \\ 5 & -13 & -4 & -7 \\ -1 & 4 & 1 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix}$.

06) Let $A = \begin{pmatrix} a & 1 & 2 \\ 1 & 1 & 1 \\ -1 & 1 & 1-a \end{pmatrix}$, where a is a real number.
Find the rank of A.

07) Let $A = \begin{pmatrix} 2 & 1 & -3 & -6 \\ 3 & -3 & 1 & 2 \\ 1 & 1 & 1 & 2 \end{pmatrix}$. Find non-singular matrices P and Q such that PAQ is a normal form.

08) Let $A = \begin{pmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{pmatrix}$. Find non-singular matrices P and Q such that $PAQ = I$.
Hence, find A^{-1}.

09) Consider the following matrices:

(a) $A = \begin{pmatrix} 2 & 2 & -1 \\ 4 & 0 & 2 \\ 0 & 6 & -3 \end{pmatrix}$ (b) $B = \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 2 & 4 & 2 \\ 0 & 2 & 2 & 1 \end{pmatrix}$

(c) $C = \begin{pmatrix} 1 & 2 & 1 & -1 \\ 9 & 5 & 2 & 2 \\ 7 & 1 & 0 & 4 \end{pmatrix}$ (d) $B = \begin{pmatrix} 1 & 0 & -2 & 1 & 0 \\ 0 & -1 & -3 & 1 & 3 \\ -2 & -1 & 1 & -1 & 3 \\ 0 & 3 & 9 & 0 & -12 \end{pmatrix}$

(i) Find the row echelon form of the above matrices.
(ii) Write down the basic columns of each matrices.
(iii) Determine the rank of the above matrices.

10) Consider the following matrices:

(a) $A = \begin{pmatrix} 1 & 2 & 1 & 1 \\ 2 & 4 & 2 & 2 \\ 3 & 6 & 3 & 4 \end{pmatrix}$ (b) $B = \begin{pmatrix} 1 & 2 & 0 & 5 \\ 2 & 3 & 1 & 4 \\ -1 & -1 & -1 & 1 \end{pmatrix}$

(i) Reduce each of the above matrices into a row echelon form.
(ii) Reduce above matrices further to its reduce row echelon form.

11)

(a) Show that the system of linear equations

$$\begin{aligned} 2x + y + 3z &= a \\ 2x + 6y + 8z &= b \\ 6x + 8y + 14z &= c \end{aligned}$$

has no solutions unless $c = 2a + b$.

(b) Determine the values of α and β for which the following system of linear equations have

(i) no solutions,
(ii) a unique solution.
(iii) infinitely many

$$\begin{aligned} 3x + y - 6z &= -10, \\ 2x + y - 5z &= -8, \\ 6x - 3y + \alpha z &= \beta. \end{aligned}$$

12) Solve the following system of linear equations using Gaussian Elimination Method:

(a)
$$\begin{aligned} x + 2y + z &= 0 \\ 2x + 3y + 3z &= 10 \\ 3x - y + 2z &= 14 \end{aligned}$$

(b)
$$\begin{aligned} x - 2y + z &= 0 \\ 2x - 7y - 3z &= 5 \\ 4x - 7y + z &= -1 \end{aligned}$$

13) Solve the following system of linear equations using Jordan-Gaussian Elimination Method:

(a)
$$\begin{aligned} 2x_2 + 3x_3 - 4x_4 &= 1 \\ 2x_3 + 3x_4 &= 4 \\ 2x_1 + 2x_2 - 5x_3 + 2x_4 &= 4 \\ 2x_1 - 6x_3 + 9x_4 &= 7 \end{aligned}$$

(b)
$$\begin{aligned} x + 2y - 3z &= 2 \\ 6x + 3y - 9z &= 6 \\ 7x + 14y - 21z &= 13 \end{aligned}$$